

Microsoft Defender for Endpoint

Master Class

Trainer DI Thomas Schleich

Version October 2026



Module 4

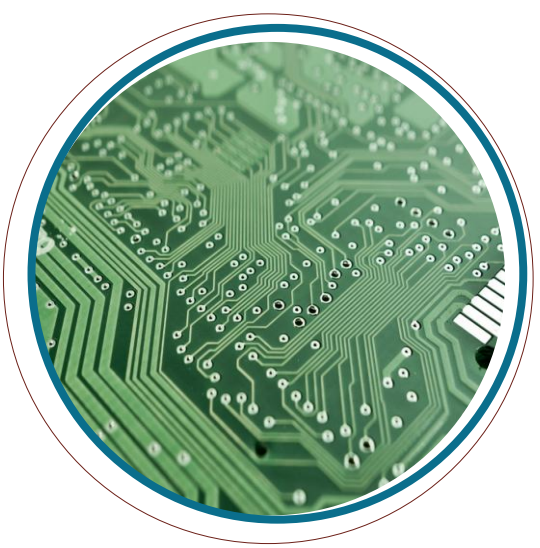
Endpoint Detection and Response

Module 4 Contents:

- Detections (Alerts, Incidents)
- Automated Investigation and Response (AIR)
- Remediation actions
- Response actions on devices

© 2023 Fast Lane

3



Detection

© 2023 Fast Lane

4

Detections

Alerts

- Organize queue
- Review, manage, investigate
- Investigate
  - Device
  - User/Identity
  - Files
  - IPs

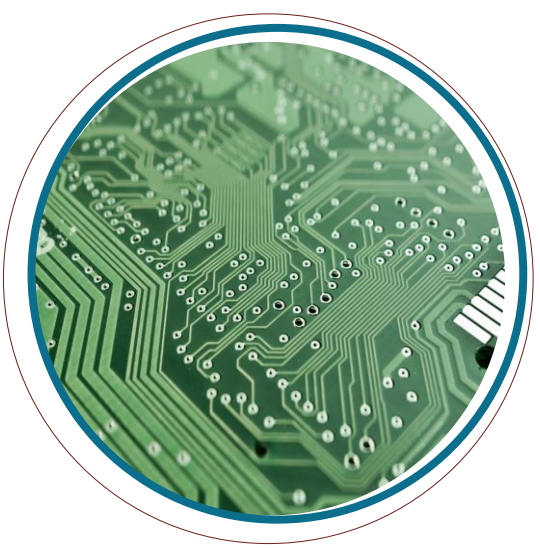
Incidents

- Organize queue
- Manage
- Investigate

Cases

- more sophisticated incident management

© 2023 Fast Lane



Automated Investigation

© 2023 Fast Lane

Source: https://learn.microsoft.com/en-us/defender-endpoint/automated-investigations

Automated Investigation and Response (AIR)

'AIR capabilities are designed to examine alerts and take immediate action to resolve breaches. AIR capabilities significantly reduce alert volume, allowing security operations to focus on more sophisticated threats and other high-value initiatives.'

- Remediation actions tracked in Action Center
- available with license
  - MDE P2
  - MDfB

© 2023 Fast Lane

AIR - Process

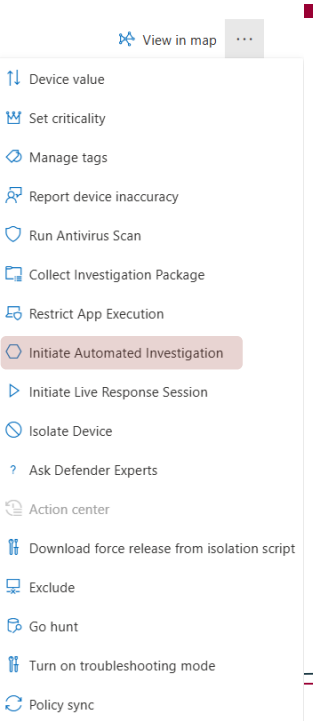


- If alert is triggered, AIR starts
- For all evidences, a verdict is generated
- Remediation actions will be executed
  - automatically or
  - upon approval

© 2023 Fast Lane

AIR - Initiate manually

- Navigate to a device page
- open Device Action menu
- click 'Initiate Automated Investigation'



© 2023 Fast Lane

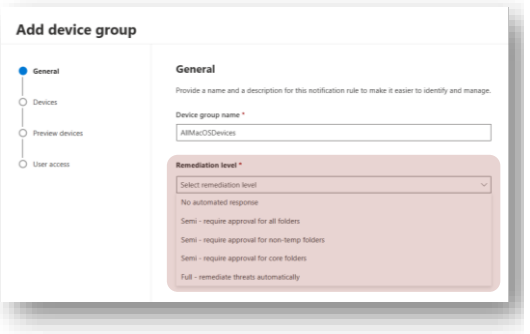
Remediation Actions

- Quarantine a file
- Remove a registry key
- Kill a process
- Stop a service
- Disable a driver
- Remove a scheduled task

© 2023 Fast Lane

Source: https://learn.microsoft.com/en-us/defender-endpoint/automation-levels

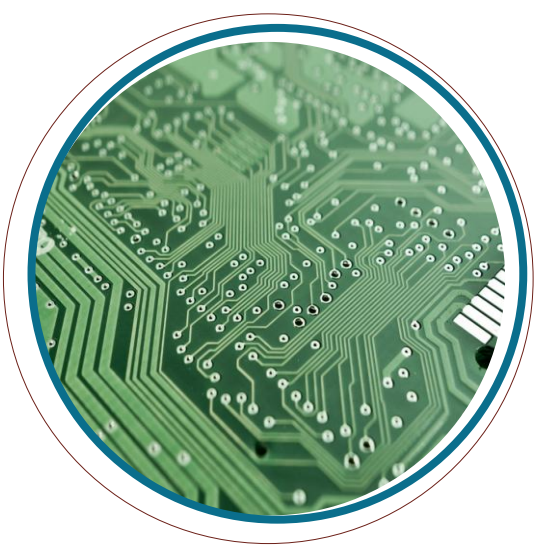
AIR - Automation Levels



- only in MDE P2 available
- set for Device Groups
- Levels
  - Full - remediate threats automatically
  - Semi - require approval for all folders
  - Semi - require approval for core folders
  - Semi - require approval for non-temp folders
  - No automated response

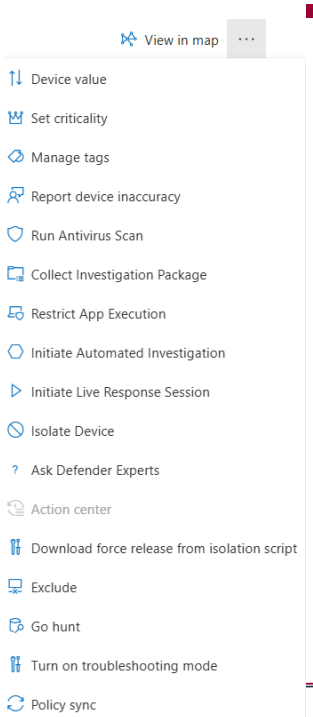
```
\users\*\appdata\local\temp\*
\documents and settings\*\local settings\temp\*
\documents and settings\*\local settings\temporary\*
\windows\temp\*
\users\*\downloads\*
\program files\
\program files (x86)\*
\documents and settings\*\users\*
```

© 2023 Fast Lane



Response Actions

© 2023 Fast Lane

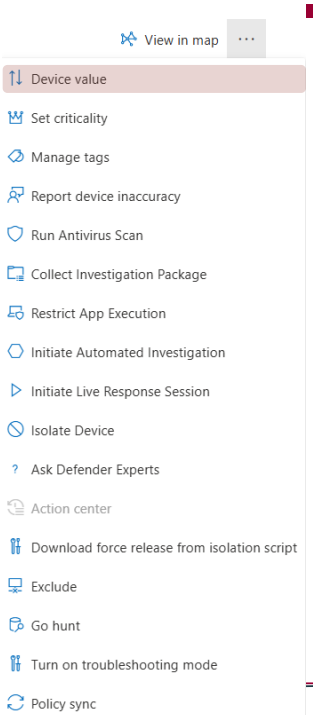


Response Actions

- on top of specific device page

- MDE P1 only:
  - Run antivirus scan
  - Isolate device

13



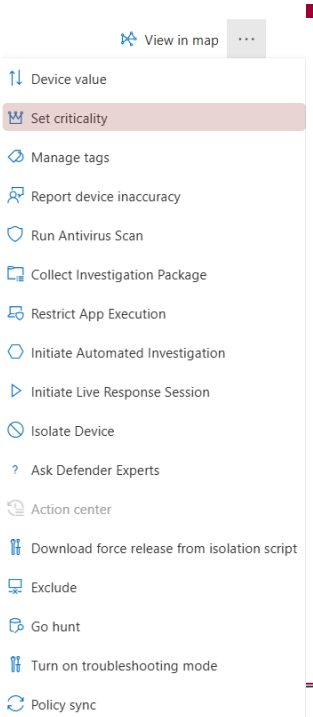
Response Actions

Device value

- *'The device value is used to incorporate the risk appetite of an individual asset into the Defender Vulnerability Management exposure score calculation.'*
- Values: Low - Normal - High
- Exposure score:

  'reflects how vulnerable your organization is to cybersecurity threats. Low exposure score means your devices are less vulnerable to exploitation.'
  - the lower the better …

14

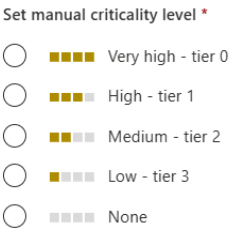


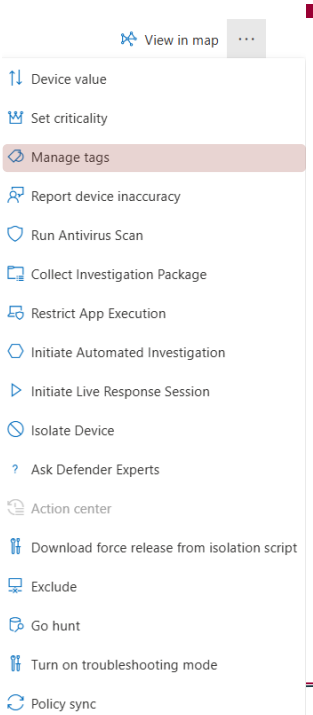
Source: https://learn.microsoft.com/en-us/security-exposure-management/critical-asset-management

Response Actions

Set criticality - part of exposure management

- *'You can prioritize security investigations, posture recommendations, and remediation steps to focus on critical assets first.'*
- Help to filter for critical devices
- used in attack path analysis
- Predefined levels for devices used - refer to source
- Values



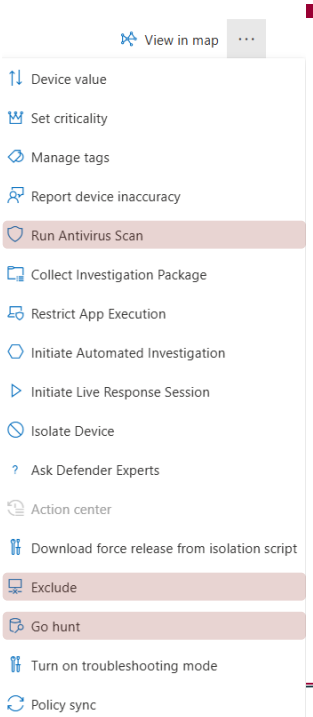


Source: https://learn.microsoft.com/en-us/defender-endpoint/machine-tags#add-device-tags-by-setting-a-registry-key-value

Response Actions

Manage tags

- additional custom information for devices
- useful for filtering device list
- useful for indication of devices (MDE-Management)
- can also be set by Registry
  - HKLM\SOFTWARE\Policies\Microsoft\Windows Advanced Threat Protection\DeviceTagging
  - Group (Reg_SZ)
- Types
  - Manual
  - Dynamic: Settings | MS Defender XDR | Asset rule mgmt
  - System (like 'Internet facing')



Response Actions

Run Antivirus Scan

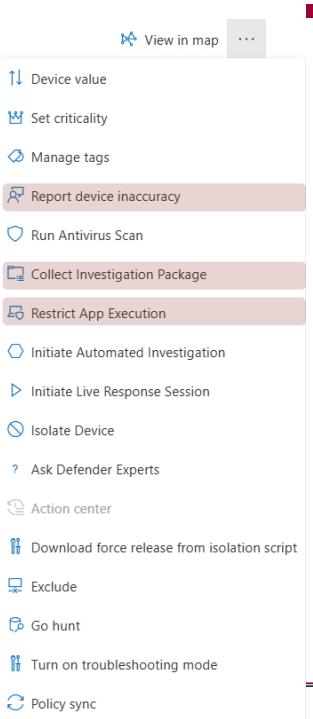
- Select a quick of full scan
- provide a comment
- see the result in Action center

Exclude

- Sets device to excluded and could be filtered out in device list
- necessary for removed or renamed devices

Go hunt

- Creates a KQL query and opens an editor with it
- for further investigation



Response Actions

Report device inaccuracy

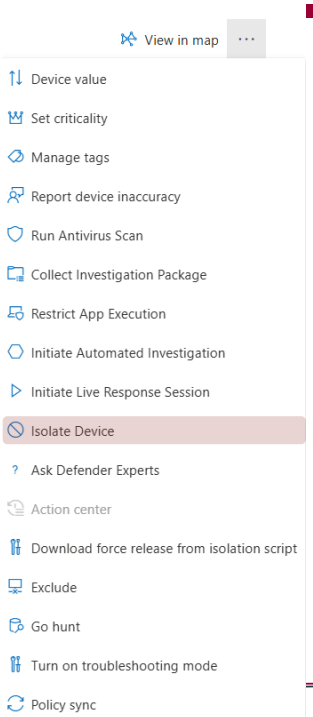
- Report inaccurate device information to Microsoft

Collect Investigation Package

- Starts the collection of important information
- Download result from Action center

Restrict App Execution

- Only apps signed by Microsoft can be started on device
- Click 'Remove App Restriction' to allow all apps again



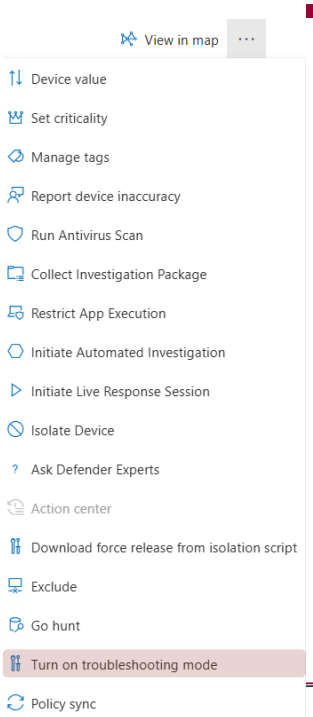
Response Actions

Isolate Device

- disconnects device from network
- connectivity to MDE retained
- Outlook, Teams and Skype communication optionally allowed
- User gets message 'Network disabled'

Enable Network on device again

- Click 'Release from isolation' in device action menu
- or: Download force release from isolation script
  - must be put to device and executed as administrator
  - a registry entry is created to start unisolation



Response Actions

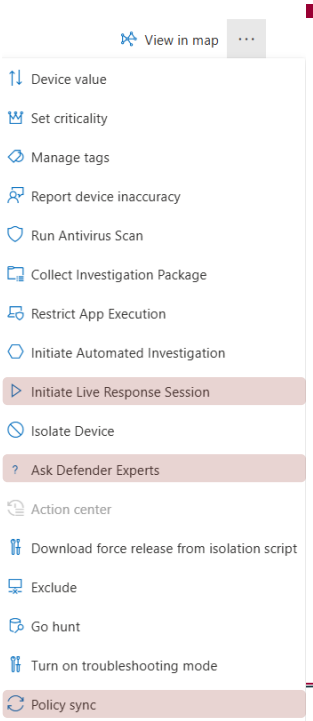
Troubleshooting mode

- Tamper protection is on by default, so no changes to the Windows Defender settings are possible
- but, with Troubleshooting mode turned on, an administrator is allowed to turn of Tamper protection on a device.

```
Set-MpPreference -DisableTamperProtection $true
```

Examples could be found here

- Unable to install application
- Microsoft Office plugin blocked by ASR



Response Actions

Ask Defender Experts
- Start talking with a human expert of Microsoft

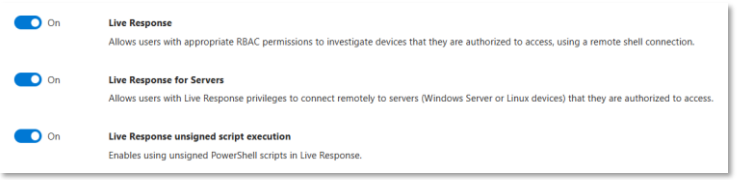
Policy sync
- starts sync of new or change Endpoint Security Policies

Initiate Live Response Session
- continued on next slides

Live Response

'Live response gives security operations teams instantaneous access to a device (also referred to as a machine) using a remote shell connection.'

Requirements
- Device OS: Windows 10, 11, Server, Linux, macOS
- Enabling: Settings | Endpoint | Adv Features
- RBAC Permissions



© 2023 Fast Lane

Live Response

- Initiated in XDR portal as device action

Commands - Basic →

cd	persistence
connect	processes
persistence	registry
processes	scheduledtasks
registry	services
scheduledtasks	startupfolders
services	status
startupfolders	trace
status	jobs
trace	

Commands - Adv ↓

analyze	library
collect	putfile
isolate	remediate
release	*scan*
run	undo

```
C:\> help_
```

```
C:\> help services_
```

© 2023 Fast Lane

Live Response

Demo

- Using help
- Find a file
- Get file from device
- Put file to device
- Run a script from library
- Run a script from library with parameters
- Clean up library
- Get registry entries
- Redirection

© 2023 Fast Lane



End of Module 4

© 2023 Fast Lane